

Why Campylobacter infections peak in January, not only in summer

Joe Martin
jomartin@ethz.ch

Campylobacteriosis in Switzerland and Liechtenstein, January 2013 – August 2026
103,485 reported cases • 164 months • FOPH reporting data and MeteoSwiss temperatures

Conclusion. January has 78% more reported cases than its cold temperature predicts; salmonellosis, used as a comparison, shows a much smaller excess. This fits the known link to meat fondue over the holidays [1]. Warmer-than-usual months go along with a few percent more cases.

MODEL ESTIMATE · FESTIVE PEAK

× 1.78

more cases in January than expected from its temperature and the year's level.
95% CI 1.63–1.94. December × 1.50 (1.32–1.70)

MODEL ESTIMATE · COMPARISON

× 1.19

for salmonellosis in January, which is not known to be linked to fondue.
95% CI 1.02–1.38; December × 0.94. Campylobacter excess 1.49 times larger (1.25–1.78)

FIGURE D · TEMPERATURE

+3.8%

cases per 1 °C warmer than usual for the time of year (same and previous month).
95% CI +1.4 to +6.3%. Salmonellosis +1.6% (–1.1 to +4.3%)

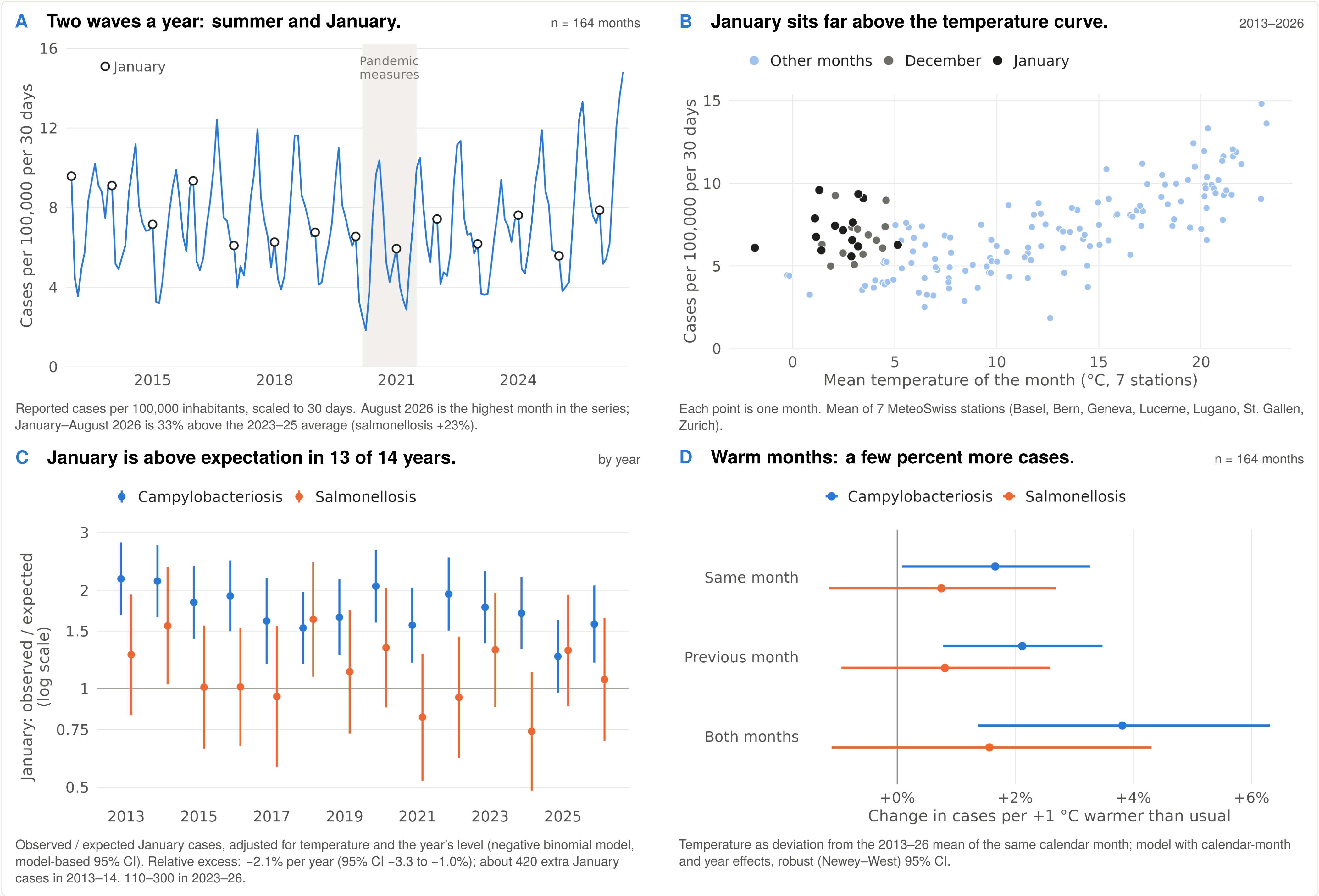
What this means

Campylobacter has **two seasons**: a summer wave that goes along with temperature, and a **holiday peak** that temperature does not account for. The January excess has become smaller since 2013 but has not disappeared. Separate boards and plates for raw and cooked meat at fondue remain a simple prevention message.

What these data cannot show

- **The cause of single cases.** Monthly national counts, no food histories; fondue is the explanation from [1], not tested here.
- **Infections, only reports.** Testing and reporting habits shape the counts.
- **Weeks.** Monthly data blur the exact timing.

FOUR VIEWS OF THE EVIDENCE



Background

Campylobacteriosis has been the most common notifiable food-associated infection in Switzerland since 1995 [2], mostly from chicken meat. A case–control study linked the winter peak to meat fondue (fondue chinoise) over Christmas and New Year [1]; a time-series study for 2008–12 found the same peak [2]. This project asks whether the peak persists, how large it is, and how temperature shapes the rest of the year.

Data	
Campylobacteriosis cases, Jan 2013–Aug 2026	103,485
Salmonellosis cases (comparison)	22,799
Months	164
MeteoSwiss climate stations	7
FOPH Infectious Diseases Dashboard (mandatory reporting, CH and FL), release of 9 September 2026. MeteoSwiss homogeneous monthly means.	

Methods & limitations

Negative binomial regression with population offset; robust standard errors for autocorrelation.

- **Temperature effect:** +2.2 to +4.9% in nine sensitivity analyses; weakest without Dec/Jan (CI includes 0).
- **January estimate:** × 1.71 to × 1.78 in four model variants; compared mainly with February.
- **Salmonellosis** shows a small, model-dependent January excess: part of the signal may be testing and reporting after the holidays.